

Supplemental Information for “BriGHT: transcriptome-regularized multimodal neuroimaging for brain disorder prediction”

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S1 Supplementary Methods

S1.1 Transcriptomic data acquisition and processing

Brain-wide transcriptomic data were obtained from the Allen Human Brain Atlas (AHBA, human.brain-map.org), which contains more than 58,000 probes sampled from 3,702 brain tissue samples from six donors. We employed the *abagen* toolbox¹ for preprocessing and mapped gene expression profiles to 116 brain regions defined by the AAL atlas, resulting in 15,633 genes. To focus on disorder-relevant molecular signals, we used large-scale meta-GWAS summary statistics for AD, ADHD, and MDD. Specifically, we utilized summary statistics from Alzheimer’s disease (AD)², ADHD³, and MDD-related GWAS studies⁴. Gene-level p -values were computed from SNP-level statistics using MAGMA⁵, and genes with nominal significance ($p < 0.05$) were retained. This resulted in 1,216 genes for AD, 1,524 genes for ADHD, and 2,463 genes for MDD, all mapped to the same 116 ROIs. These disease-specific gene sets were subsequently used to construct disorder-specific transcriptome-derived ROI hypergraphs.

S1.2 Neuroimaging data acquisition and preprocessing

Neuroimaging data were obtained from three brain disorder cohorts, including ADNI, ADHD-200, and REST-meta-MDD. Four imaging modalities were used across these cohorts: AV45-PET, FDG-PET, VBM-based structural MRI (VBM-sMRI), and resting-state fMRI (rs-fMRI).

For ADNI, preprocessed AV45-PET and FDG-PET scans were obtained from the ADNI database. Standard PET preprocessing included motion correction, spatial normalization, smoothing, and intensity normalization^{6,7}. AV45-PET images were normalized to cerebellar gray matter, and regional SUVR values were extracted using the AAL atlas and MarsBaR toolbox⁸. FDG-PET scans were aligned to subject-specific T1-weighted MRI and normalized to MNI space, with regional glucose metabolism values extracted from AAL ROIs⁹.

Structural MRI data were processed using voxel-based morphometry implemented in SPM12. T1-weighted images were normalized to MNI space, resampled, and smoothed with a Gaussian kernel. Gray matter density values were extracted from AAL ROIs and used as VBM-sMRI regional features⁸. The same VBM preprocessing strategy was applied to the cohorts where structural MRI was used.

Resting-state fMRI data from ADHD-200 and REST-meta-MDD were preprocessed using DPABI¹⁰. The preprocessing steps included removal of the first 10 volumes, slice timing correction, motion correction, spatial normalization, and smoothing. Subjects with excessive head motion were excluded. ROI-wise time series were extracted, and functional connectivity matrices were constructed using Pearson correlations between ROI time series. These correlation matrices were thresholded to obtain binary adjacency matrices. To construct compact node features, principal component analysis was applied to ROI-wise temporal signals, and the first principal component was retained as the representative feature. This choice was motivated by prior studies showing that dominant low-rank components can capture discriminative information^{11,12}. As shown in Table S1, using the first principal component provided the best performance in this setting while reducing computational cost; therefore, this setting was used in all experiments.

Table S1: PCA component selections for fMRI ROI graph construction.

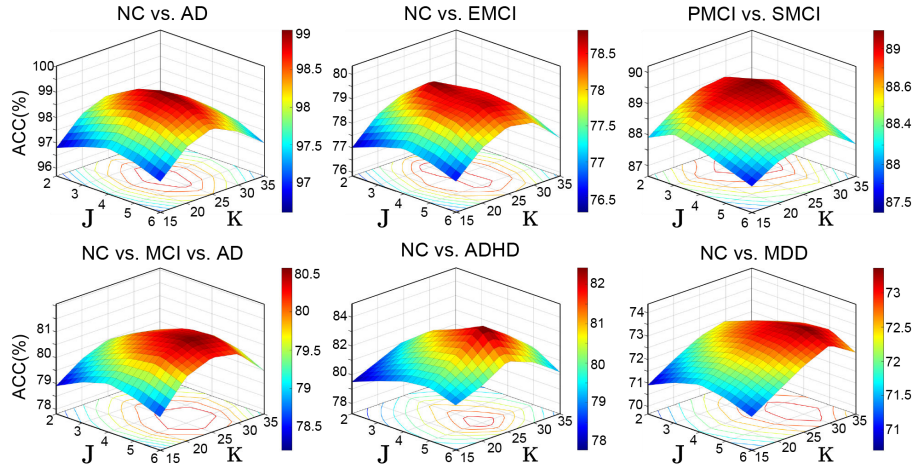
Number of PCs	NC vs. ADHD			
	ACC (%)	F1 (%)	AUC (%)	Time (s/epoch)
1	80.2±1.3	79.3±1.4	80.1±1.8	0.54
10	74.1±1.3	72.7±1.6	78.7±1.1	1.08
20	77.5±1.5	76.3±1.6	75.7±1.3	1.40
50	75.8±2.1	75.8±1.8	72.2±1.9	1.61

S2 Supplemental Results

S2.1 Hyperparameter sensitivity analysis

S2.1.1 Number of modules J and selected ROIs K

We assessed the sensitivity of BriGHT to key hyperparameters used in hypergraph construction, including the number of co-expression modules J and the number of selected ROIs per hyper-edge K . As shown in Fig. S1, BriGHT maintains stable performance across a broad range of parameter settings, and the selected configuration lies within a high-performing region. These results indicate that the framework is robust to hyperparameter choices.

**Figure S1:** Hyperparameter analysis of J and K for hypergraph construction.

S2.1.2 p -Laplacian orders

We further examined the sensitivity of BriGHT to the p -Laplacian orders used in layer-wise hypergraph regularization. Specifically, we varied p_1 and p_2 for the first and second graph attention layers over a broad range and evaluated performance across all prediction tasks. As shown in Fig. S2, intermediate p values generally perform well, with the best results primarily obtained when p lies in an intermediate range. These results suggest that allowing flexible p -Laplacian smoothing can help balance transcriptomic structural regularization and subject-specific neuroimaging variation.

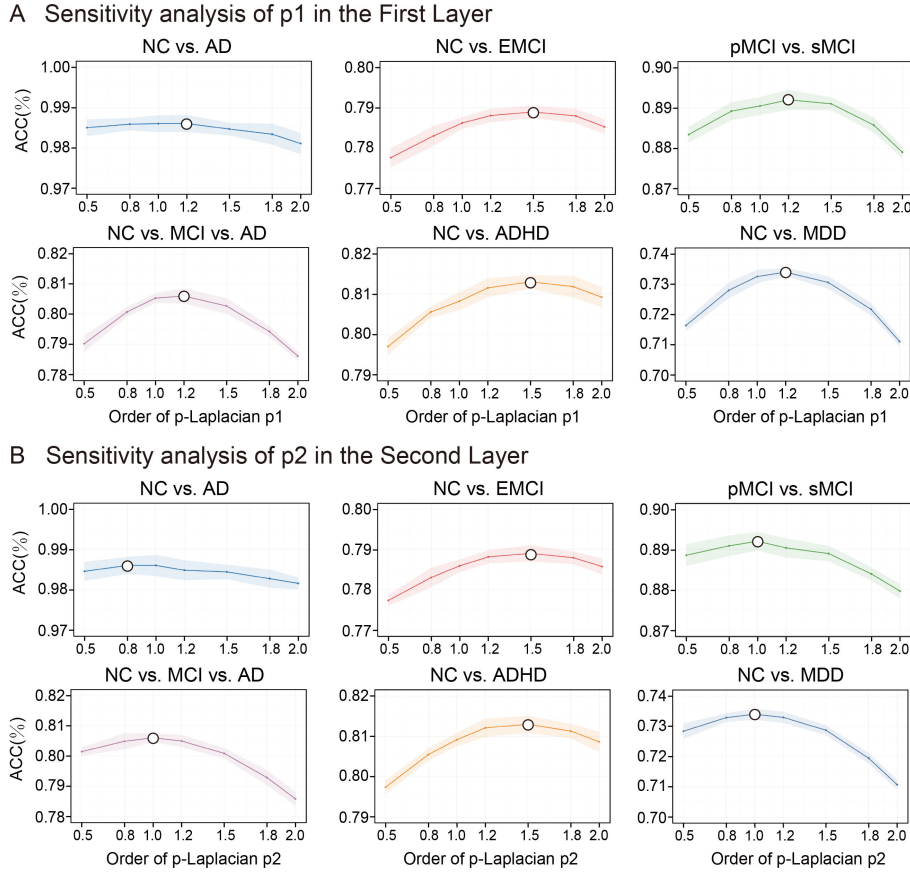


Figure S2: Hyperparameter analysis of the p -Laplacian orders (p_1, p_2). Solid lines denote mean performance across 5-fold cross-validation, and shaded regions indicate standard error. Best configurations are marked with white dots.

S2.1.3 Regularization weight λ

We also evaluated the sensitivity of BriGHT to the regularization weight λ , which balances the classification loss and the transcriptomic hypergraph regularization term. As shown in Fig. S3, performance remains stable across the range of λ values, while overly large values lead to mild degradation in some tasks. This suggests that moderate transcriptomic anchoring is sufficient to guide ROI-level representation learning.

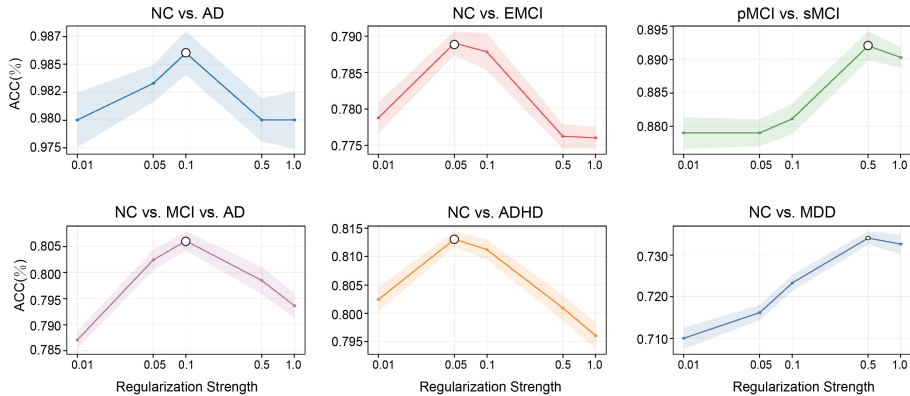


Figure S3: Hyperparameter analysis of the regularization strength λ .

S2.2 Representation robustness analysis

Representation visualization. We visualized representations from three stages: the original imaging features, the representations after transcriptome-regularized graph learning, and the fused multimodal embeddings. As shown in Fig. S4A, diagnostic groups substantially overlap

in the original feature space, but become progressively more separable after graph learning and fusion. This suggests that the proposed modules improve the discriminative organization of the latent space.

Robustness analysis. We further evaluated robustness by randomly masking different proportions of input features and comparing the resulting performance degradation. As shown in Fig. S4B, performance decreases as the masking ratio increases, but BriGHT maintains higher accuracy than competing methods across masking levels and tasks. This indicates that the representations learned from our method are less sensitive to feature-level perturbations, consistent with the stabilizing effect of the proposed regularization and fusion modules.

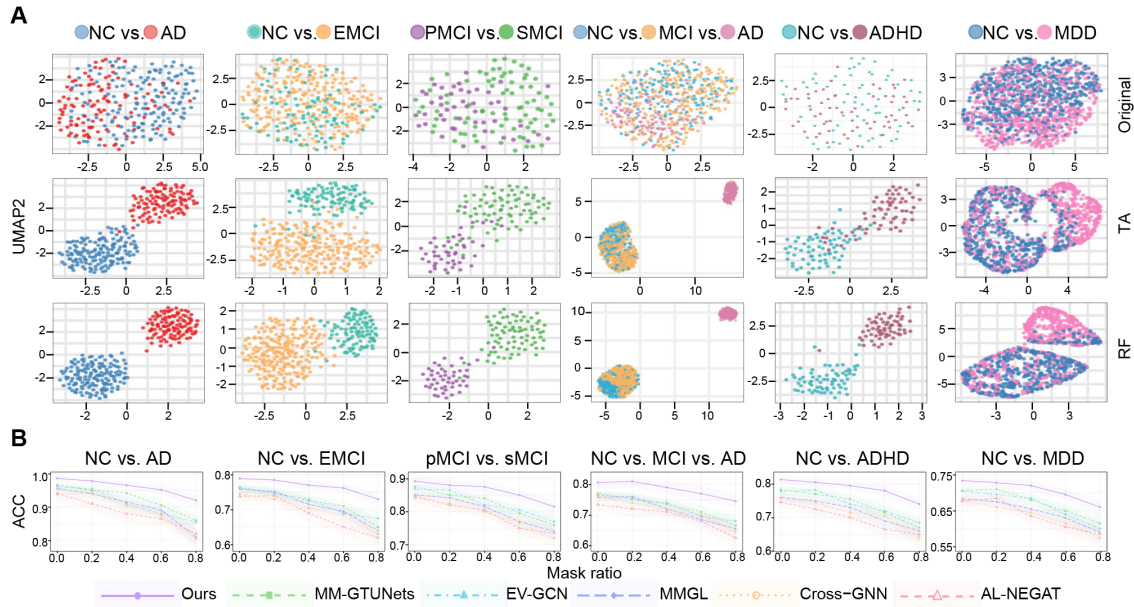


Figure S4: Representation and robustness analysis. (A) UMAP visualization of original features, transcriptome-regularized graph representations, and fused multimodal embeddings. (B) Robustness evaluation under increasing feature masking ratios.

S2.3 Supplementary biological interpretability analysis

Fig. S5 provides additional visualization of the top ROIs identified by feature ablation. Panel A shows the top-10 ROIs for each prediction task, and Panel B presents the corresponding NeuroQuery-derived functional relevance maps. These results complement the main-text interpretability analysis by showing that the identified regions exhibit task-specific spatial patterns and overlap with literature-supported functional relevance maps.

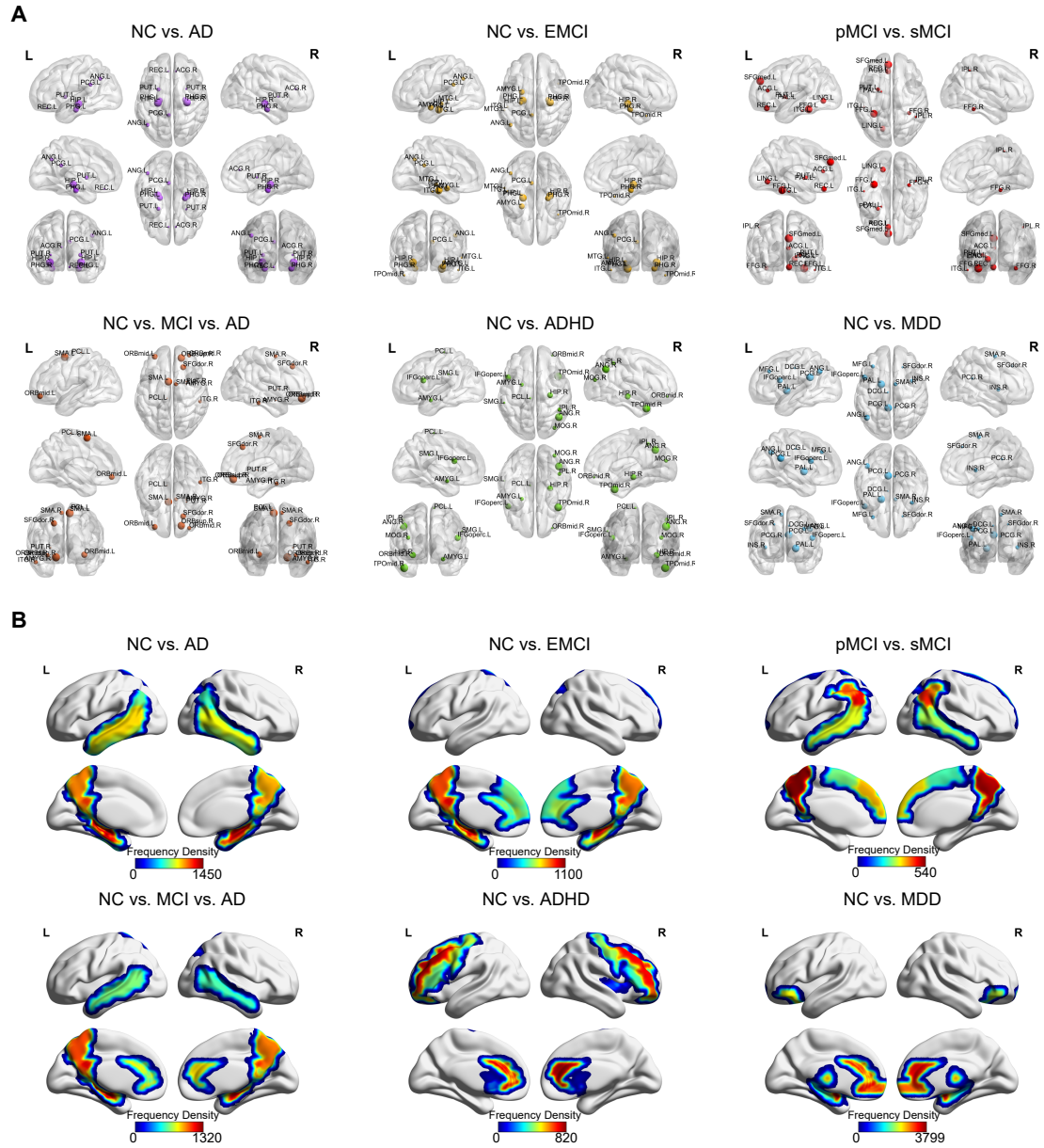


Figure S5: Discriminative brain ROIs and their functional relevance. (A) Top 10 discriminative regions and (B) their literature-derived functional relevance from NeuroQuery meta-analysis.

S2.4 Pathway enrichment analysis of disease-associated genes

We performed GO and KEGG pathway enrichment analyses to assess the biological relevance of the disease-associated genes used for hypergraph construction. As shown in Fig. S6, AD-associated genes were enriched in immune-related signaling, synaptic organization, and hormone-related pathways; ADHD-associated genes were mainly related to neurodevelopmental and synaptic processes; and MDD-associated genes were enriched in synapse-related processes, neuronal signaling, and glutamatergic transmission. These results are broadly consistent with known biological mechanisms of the corresponding disorders and support the use of these gene sets for constructing transcriptome-derived structural references.

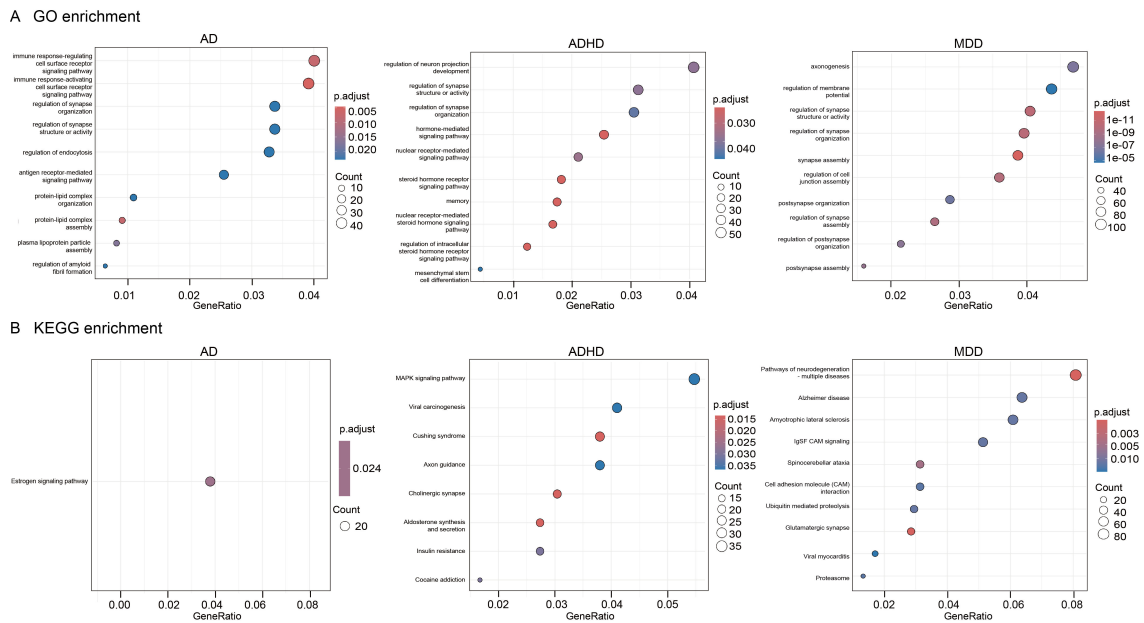


Figure S6: Pathway enrichment analysis of disease-associated genes used for hypergraph construction. (A) GO enrichment results for AD, ADHD, and MDD. (B) KEGG pathway enrichment results for AD, ADHD, and MDD.

Supplemental Reference

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